



Digital SLRs use larger CCD sensors than point and shoot compact digital cameras

The Bayer Pattern

To understand the breakthrough, one needs first to know the mechanics of a sensor itself, be it CMOS or CCD. The sensor is an array of millions of sensor pixels, which collect and store photons from light captured by the camera's lens.

By themselves, these sensor pixels would be able to capture only greyscale images. To capture colour, a filter needs to be placed over each pixel, thus letting each pixel capture only a particular colour from the incoming photons. The pixels, of course, are restricted to RGB (red, green and blue).

Once the pixels capture these colours, smart algorithms take over – in a simplistic explanation, they make sense of the colours from the millions of pixels on a sensor, join the dots, so to speak, and deliver the best image.

In 1976, Kodak scientist Bryce Bayer figured out that the best output requires the best array of filters. Experiments led him to conclude that this optimal array was an alternating series of pixel filters, with the first line being red-green and the next one being green-blue, such that green becomes the dominant filter colour on the sensor (see Fig 1).

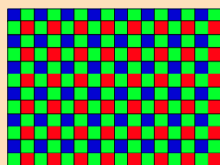


Fig 1

The Bayer Pattern has since become the basis for creating sensors, be it for CCD, CMOS, or any other. It is so widely used, that only an improvement of the same technology makes sense in terms of manufacturing, as an overhaul of the system is too much of a cost to bear.

wavelengths, and is designed to absorb light. The lack of filtering light to detect colour information makes the sensors more sensitive, thus providing greater luminescence, which translates to better clarity in the end result.

In the new array, the panchromatic pixels would be predominant. So, (fig 2) the panchromatic pixels would be twice in number those of green pixels, while the green to blue/red ratio would still be 2:1. What's more, the array could use different patterns with these new pixels, depending on the kind of applications

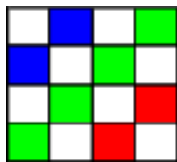


Fig 2



Applications

The two main applications for this new sensor pattern are clicking photos in dim lighting, or snapping a picture of moving objects (see comparison shot provided by Kodak).



Technology has improved image stabilisation

In consumer technology, the obvious advantage lies in mobile phone cameras. Low-light shots are still an eyesore, even with the dual LED flash on some of the new handsets. Sensitivity to light is the key to delivering less grainy (i.e. less noisy) images, and that's exactly what panchromatic pixels offer.

As for blurry shots, mobile phones as well as end-user digital cameras have fallen prey to moving hands and faces. The promise to get rid of these, without the added hassle of image stabilisation algorithms, is quite attractive, as it reduces processing power and bloated software.

While Kodak is currently working on CMOS versions of the panchromatic array, and already has a sample sensor ready, the CCD version is more promising for scientific institutions, especially those involved in space science.

It's pure light out there, and in the near future, we might just get to see the most fantastic shots of the universe!

and power requirements that the sensor demands.

"This represents a new generation of image sensor technology and addresses one of the great challenges facing our industry – how to capture crisp, clear digital images in a poorly lit environment," said Chris McNiffe, General Manager of Kodak's Image Sensor Solutions group.

Distilling distortion

While the issue with photo filters should be addressed in the near future, US-based scientists from two universities are looking a bit further down the years. After all, they're going after a much more complex problem, at least in terms of pure hardware engineering: the distortion caused by the relationship between flat photodetectors and curved lenses.

University of Wisconsin-Madison's Electrical and Computer Engineering Associate Professor Zhenqiang "Jack" Ma and student Hao-Chih Yuan have developed a flexible light-sensitive material that, they say, could revolutionise photography. The team also includes UW-Madison Materials Science and Engineering Professor Max Lagally and University of Michigan Professor Pallab Bhattacharya.

Their inspiration for this potentially ground-breaking technology: the human eye! Light enters the eye through a single lens, but at the back of the eye, the image falls upon the curved retina, eliminating distortion. "If you can make a curved imaging plane, you just need one lens," said Ma. "That's why our development is extremely important."

Ma and his group have created curved photodetectors with specially fabricated nanomembranes – extremely thin, flexible sheets of germanium, a very light-sensitive material often used in high-end imaging sensors. The researchers can then apply the nanomembranes to any polymer substrate, such as a thin, flexible piece of plastic. Currently, the group has demonstrated photodetectors curved in one direction, but Ma hopes to develop hemispherical sensors soon.

"We can easily realise very high-density flexible and sensitive imaging arrays, because the photodetector material germanium itself is extremely bendable and extremely efficient in absorbing light," Ma said.

And what next?

Present research, near-future technologies and distant-future possibilities aside, what's next in the big picture for sensor technologies?

The answer may lie with a bunch of Harvard guys, who have created what they call "black silicon". This material can absorb more light than any other commercial material on the planet, so be on the lookout for great things from a spin-off venture called SiOnyx.

After all, a material that captures the infrared spectrum is bound to capture the world's attention with it! ■

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Why distortion occurs

Ever taken a photo of a person from up close? Some can turn out looking like images from an amusement park's 'Hall of Mirrors', with an enlarged nose or a hand as big as a head.

This distortion happens because of the distance between the lens and the photodetector – a light-sensitive material such as the sensor in a digital camera.

When a camera records an image, light passes through the lens and lands on the photodetector. However, a lens bends the light and curves the focusing plane.

In a digital camera, the point where the focusing plane meets the farther it is from that central focus point.

High-end digital cameras correct this problem by incorporating multiple panes of glass to refract light and flatten the focusing plane.

However, such lens systems – like the mammoth telephoto lenses sports photographers use – are large, bulky and expensive. And even high-quality lenses stretch the edges of an image somewhat, thus never providing a "true image" for the photo-enthusiast.